

RIB FRACTURE DETECTION USING DEEP LEARNING ON CHEST X-RAY IMAGES

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1. INTRODUCTION

Rib fractures generally result from trauma but may be from pathologic fracture of a metastatic lesion. Due to their subtle appearance, acute fractures are often missed on radiographs [1]. With demineralization of ribs in older patients, detecting such fractures from only minimal trauma is more challenging. An automated method of diagnosing rib fractures would be highly desirable. Our goal was to investigate the use of deep learning models to detect rib fractures in chest X-rays in both trauma events and incidental rib fracture detection.

2. METHODOLOGY

2.1. Deep Learning Model Selection

We assessed the performance ResNet50 and EfficientNet v2 models in classifying X-ray images as containing a rib fracture or otherwise, using binary classification validation set accuracy as the key evaluation metric. The ResNet50 model was used as the baseline architecture [2] which introduced training residuals to resolve the problem of accuracy saturation caused by the stacking of additional layers in a deep learning model. We chose EfficientNet as the second model architecture [3] and tested three distinct models, each with different input sizes (B0 with 224x224 pixels, B2 with 260x260 pixels, and B4 with 380x380 pixels).

2.2. Dataset Overview

Images were sourced from a dataset of chest X-ray images from radiographic studies at Beth Israel Deaconess Medical Center in Boston, MA. 2,273 AP chest X-ray images were used in our research across 1,482 unique patients, with 1,325 images labeled as positive for rib fracture(s) and 948 labeled as negative for any fracture. The ages of patients in our input data range from 18 – 98, with a mean of 62 ± 18.7 [4-6].

2.3. Model Training and Evaluation

We split the full set of input images (2,273 images) into a training set (80%), validation set (10%), and test set (10%). ResNet50 and the EfficientNet models are pre-trained on the ImageNet dataset. We trained the models for 10 epochs using a categorical cross-entropy loss function and Adam optimizer with a default learning rate of 0.001. For the EfficientNet models, we employed fine-tuning section by unfreezing the last 10 feature extraction layers, creating more trainable parameters.

3. RESULTS

Training set and validation set accuracy and AUC were recorded and analyzed. For each model, the epoch producing the greatest validation accuracy was regarded as the model

version to evaluate. After training these models on all images in the dataset, EfficientNet (model B2) performed best with a validation set accuracy of 82.14%. It also had a validation AUC (area under the receiver operator curve) of 86.29%, compared to ResNet50 which produced validation accuracy of 75.45% and validation AUC of 79.13%. EfficientNet B4 produced training accuracy of 81.81% and training AUC of 89.73%. Across all four models, validation accuracy ranged from 70%-82% and validation AUC ranged from 75%-86%.

4. DISCUSSION

This research has shown that classifying individual images as containing a rib fracture vs no fracture can be performed with strong accuracy by deep learning models. The accuracy of 82.14% achieved by EfficientNet v2 B2 shows that deep learning models for X-ray fracture detection are promising.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study was conducted retrospectively using open access human subject data. Ethical approval was not required.

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7. REFERENCES

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